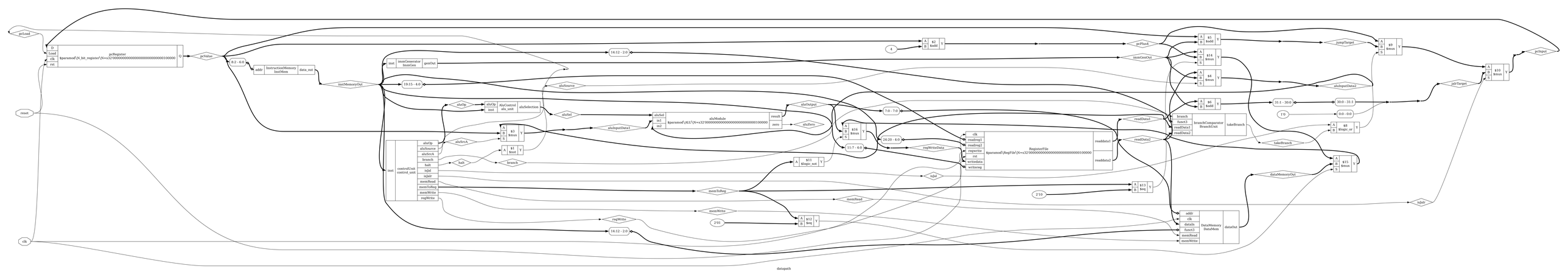

pcLoad
D
Load
clk
rst
pcRegister
$paramod\N_bit_register\N=s32'00000000000000000000000000100000
Q
pcValue
reset
clk
8:2 - 6:0
addr
InstructionMemory
InstMem
data_out
instMemoryOut
inst
immGenerator
ImmGen
genOut
19:15 - 4:0
14:12 - 2:0
aluOp
aluOp
inst
AluControl
alu_unit
aluSelection
aluSource
aluSel
A
B
S
$3
$mux
Y
aluInputData1
aluSel
in1
in2
aluModule
$paramod\ALU\N=s32'00000000000000000000000000100000
result
zero
aluOutput
aluZero
aluSrcA
A
$1
$not
Y
halt
branch
inst
controlUnit
control_unit
aluOp
aluSource
aluSrcA
branch
halt
isJal
isJalr
memRead
memToReg
memWrite
regWrite
memToReg
memRead
memWrite
regWrite
14:12 - 2:0
2'01
A
B
$12
$eq
Y
A
$11
$logic_not
Y
A
B
S
$16
$mux
Y
7:0 - 7:0
24:20 - 4:0
11:7 - 4:0
regWriteData
isJal
clk
readreg1
readreg2
regwrite
rst
writedata
writereg
RegisterFile
$paramod\RegFile\N=s32'00000000000000000000000000100000
readdata1
readdata2
readData1
readData2
4
A
B
$2
$add
Y
pcPlus4
immGenOut
A
B
$5
$add
Y
A
B
S
$14
$mux
Y
A
B
S
$4
$mux
Y
A
B
$6
$add
Y
branch
funct3
readData1
readData2
branchComparator
BranchUnit
takeBranch
takeBranch
jumpTarget
aluInputData2
31:1 - 30:0
30:0 - 31:1
1'0
0:0 - 0:0
A
B
$8
$logic_or
Y
A
B
S
$9
$mux
Y
jalrTarget
A
B
S
$10
$mux
Y
pcInput
2'10
A
B
$13
$eq
Y
dataMemoryOut
A
B
S
$15
$mux
Y
isJalr
addr
clk
dataIn
funct3
memRead
memWrite
DataMemory
DataMem
dataOut

datapath